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#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include <NTPClient.h>
#include <WiFiUdp.h>
#include "DHT.h"

// =====
// 1. CONFIGURATION & CREDENTIALS
// =====
const char* ssid = "YOUR_WIFI_SSID"; // Replace with your Wi-Fi name
const char* password = "YOUR_WIFI_PASSWORD"; // Replace with your Wi-Fi password
const String botToken = "YOUR_BOT_TOKEN"; // Replace with your Telegram Bot Token

// =====
// 2. PINS & HARDWARE SETUP
// =====
#define DHTPIN 4 // DHT11 data pin connected to GPIO 4
#define DHTTYPE DHT11 // DHT 11
#define SOIL_PIN 34 // Analog pin for Soil Moisture Sensor

DHT dht(DHTPIN, DHTTYPE);

// =====
// 3. OBJECTS & TIMERS
// =====
WiFiClientSecure secured_client;
UniversalTelegramBot bot(botToken, secured_client);

WiFiUDP ntpUDP;
// NTP Client for Time (Offset for IST: 5 hours 30 mins = 19800 seconds)
NTPClient timeClient(ntpUDP, "pool.ntp.org", 19800, 60000);

unsigned long lastTimeBotRan = 0;
const int botRequestDelay = 1000; // Check for new messages every 1 second

unsigned long lastAlertTime = 0;
const int alertDelay = 3600000; // Only send automatic alerts once an hour to avoid spam

// Sensor thresholds based on your poster
const int MOISTURE_DRY = 25;
const int MOISTURE_WET = 80;
const float TEMP_COLD = 15.0;
const float TEMP_HOT = 33.0;

// =====
// 4. HELPER FUNCTIONS

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// =====

// Function to map analog reading to percentage (0-100%)
// Note: Dry soil = higher analog value (e.g., 4095), Wet soil = lower (e.g., 1000)
int getSoilMoisture() {
    int analogVal = analogRead(SOIL_PIN);
    int moisturePercent = map(analogVal, 4095, 1000, 0, 100);
    moisturePercent = constrain(moisturePercent, 0, 100);
    return moisturePercent;
}

// Function to generate time-based greeting
String getGreeting() {
    timeClient.update();
    int currentHour = timeClient.getHours();

    if (currentHour >= 5 && currentHour < 12) {
        return "Good morning, Kanan! 🌞";
    } else if (currentHour >= 12 && currentHour < 17) {
        return "Good afternoon, Kanan! 🌤️";
    } else if (currentHour >= 17 && currentHour < 21) {
        return "Good evening, Kanan! 🌃";
    } else {
        return "Hey Kanan, shouldn't you be asleep? 🌙";
    }
}

// Function to get current formatted Date and Time
String getDateTimeString() {
    timeClient.update();
    // NTPClient doesn't give date easily natively, but we can get formatted time
    return "Time: " + timeClient.getFormattedTime();
}

// Function to generate the Plant's Mood/Sarcastic Reply
String getPlantMood(int moisture, float temp, float hum) {
    String mood = "";

    // Random joke generator seed
    int jokeNum = random(1, 4);

    if (moisture < MOISTURE_DRY) {
        if (jokeNum == 1) mood += "I'm drier than a stand-up comedian's humor. WATER ME! 🌵\n";
        else if (jokeNum == 2) mood += "Hey Kanan, do I look like a cactus to you? Because I'm turning into one. 🌵\n";
        else mood += "Mayday! Mayday! Soil moisture critical. Send water ASAP. 🚒\n";
    } else if (moisture > MOISTURE_WET) {

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    mood += "Glug glug... Did you mistake me for a water lily? I'm drowning here! 🌊\n";
} else {
    mood += "Soil is perfectly moist. I'm feeling hydrated and vibing. ✨\n";
}

if (temp > TEMP_HOT) {
    mood += "Also, it's boiling hot! I'm practically sweating chlorophyll right now. 🥵\n";
} else if (temp < TEMP_COLD) {
    mood += "Brrr... it's freezing! Get me a tiny plant sweater, will you? 🥶\n";
}

// If everything is perfect
if (moisture >= MOISTURE_DRY && moisture <= MOISTURE_WET && temp >=
TEMP_COLD && temp <= TEMP_HOT) {
    if (jokeNum == 1) mood += "Honestly, conditions are flawless right now. 10/10 parenting,
Kanan. ❤️\n";
    else mood += "Living my best plant life today. Just photosynthesizing and chilling. 🌿\n";
}

return mood;
}

// =====
// 5. MESSAGE HANDLING
// =====
void handleNewMessages(int numNewMessages) {
    for (int i = 0; i < numNewMessages; i++) {
        String chat_id = bot.messages[i].chat_id;
        String text = bot.messages[i].text;
        String from_name = bot.messages[i].from_name;

        if (text == "/start") {
            String welcome = "Hello " + from_name + "! I am your IoT Interactive Plant. 🌱\n\n";
            welcome += "Send /status to check how I'm doing.\n";
            welcome += "Send /joke if you want to hear a bad plant pun.\n";
            bot.sendMessage(chat_id, welcome, "");
        }

        else if (text == "/status") {
            float h = dht.readHumidity();
            float t = dht.readTemperature();
            int m = getSoilMoisture();

            if (isnan(h) || isnan(t)) {
                bot.sendMessage(chat_id, "Ugh, my sensors are acting up. I can't feel my leaves.
(Check wiring!) 🛠️", "");
                return;
            }
        }
    }
}

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String response = getGreeting() + "\n\n";
response += "📊 *Here is my Health Report:* \n";
response += "🌡️ Temperature: " + String(t) + " °C \n";
response += "💧 Humidity: " + String(h) + " % \n";
response += "🌱 Soil Moisture: " + String(m) + " % \n\n";

response += "🧠 *My Mood:* \n" + getPlantMood(m, t, h) + "\n";
response += "🕒 " + getDateTimeString();

bot.sendMessage(chat_id, response, "Markdown");
}

else if (text == "/joke") {
  int r = random(1, 4);
  if (r == 1) bot.sendMessage(chat_id, "What do you call a nervous plant? \n... A sweater plant. Ba-dum-tss! 🥁", "");
  else if (r == 2) bot.sendMessage(chat_id, "Why did the plant go to therapy? \n... To get to the root of its problems. 🌱", "");
  else bot.sendMessage(chat_id, "I wet my plants! ... Wait, no, that's your job, Kanan.", "");
}

else {
  bot.sendMessage(chat_id, "I only speak a few commands, Kanan. Try /status or /start.", "");
}
}
}

// =====
// 6. MAIN SETUP
// =====

void setup() {
  Serial.begin(115200);

  // Connect to Wi-Fi
  Serial.print("Connecting to Wi-Fi");
  WiFi.begin(ssid, password);
  secured_client.setInsecure(); // Required for ESP32 to connect to Telegram without certs

  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(500);
  }
  Serial.println("\nConnected to Wi-Fi!");

  // Initialize Sensors & Time
  dht.begin();

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timeClient.begin();

// Seed the random generator for jokes
randomSeed(analogRead(33));
}

// =====
// 7. MAIN LOOP
// =====
void loop() {
  // 1. Check for incoming Telegram messages
  if (millis() - lastTimeBotRan > botRequestDelay) {
    int numNewMessages = bot.getUpdates(bot.last_message_received + 1);
    while (numNewMessages) {
      handleNewMessages(numNewMessages);
      numNewMessages = bot.getUpdates(bot.last_message_received + 1);
    }
    lastTimeBotRan = millis();
  }

  // 2. Automatic Health Monitoring (Alert System)
  if (millis() - lastAlertTime > alertDelay) {
    float t = dht.readTemperature();
    int m = getSoilMoisture();

    if (!isnan(t)) {
      bool needAlert = false;
      String alertMsg = "⚠️ *PLANT EMERGENCY!* ⚠️ \nHey Kanan, ";

      if (m < MOISTURE_DRY) {
        alertMsg += "I'm dangerously dry! (" + String(m) + "%). Water me immediately!\n";
        needAlert = true;
      } else if (m > MOISTURE_WET) {
        alertMsg += "I'm swimming in here! (" + String(m) + "%). Let me drain!\n";
        needAlert = true;
      }

      if (t > TEMP_HOT) {
        alertMsg += "It's way too hot for me (" + String(t) + "°C)\n";
        needAlert = true;
      } else if (t < TEMP_COLD) {
        alertMsg += "I'm freezing (" + String(t) + "°C)\n";
        needAlert = true;
      }

      if (needAlert) {
        // NOTE: To make auto-alerts work, you need a specific Chat ID.
        // Once you start the bot, it will reply to that chat.

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// For broad alerts, we can print to serial, or if you know your Chat ID, use:  
// bot.sendMessage("YOUR_CHAT_ID", alertMsg, "Markdown");  
Serial.println(alertMsg);  
}  
  
lastAlertTime = millis();  
}  
}  
}
```